

Chapter 1: Introduction

✅ Lessig's 4 Modalities (Law, Norms, Market, Architecture)

Used to understand how technology affects people's behaviour.

Modality	Meaning (Simple)	Example
Law	Rules made by government	PDPA protects personal data, Cyber laws punish hackers
Norms	What society thinks is acceptable	People criticize you for pirating movies
Market	Prices or economy influence behaviour	If Netflix is cheap, fewer people pirate shows
Architecture (Code)	How technology itself controls behaviour	Passwords, firewalls, or app restrictions stop misuse

💡 **Use this for analysis** – Example:
→ For *Quantum Computing*:
Law = new rules for encryption,
Norms = people's fear of privacy loss,
Market = tech companies compete,
Architecture = quantum-safe cryptography.

✅ Globalization (Good & Bad)

Good (Benefits):

1. More trade → more jobs. (e.g., Apple hires workers in many countries)
2. Easier access to technology. (e.g., Malaysia uses 5G from US/Japan tech)
3. Cultural sharing → we enjoy K-pop, anime, Western films.
4. Study or work overseas easily.

Bad (Harms):

1. The rich get richer, poor get poorer.
 2. Cheap labour in poor countries → unfair pay (e.g., factory workers in Bangladesh).
 3. Local culture slowly disappears (everyone uses English, not native languages).
 4. Brain drain → smart people leave Malaysia to work in Singapore.
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✓ Digital Divide (3 Stages)

Stage	Meaning (Simple)	Example
Access Divide	Some people can't afford devices or internet	Rural area no internet, city area has fibre
Usage Divide	Some people don't know how to use tech properly	Old people can't use e-wallets
Impact Divide	Some benefit more from tech	Company using AI earns more than manual workers

Chapter 3: Digital Entertainment

✓ Negative Effects for All Age Groups

Group	Problems	Example
Children	Copy violence, short attention, poor school results	Kid hits others after watching fighting shows
Teenagers	Game addiction, isolation, anger	Teen skips homework to play mobile games
Adults	Stress, time waste, family neglect	Parent ignores kids while binge-watching dramas
Seniors	Eye strain, confusion, loneliness	Grandma plays online bingo too long and forgets meals

✓ Physical Problems

- **Nintendonitis:** Hand pain from gaming or phone use.
- **DVT:** Sitting too long → blood clot.
- **Eye Strain:** From long screen time.
- **Sleep Problems:** Blue light keeps brain awake.
- **Headaches:** From bright or long exposure.



Example:

Gamers playing overnight → neck pain, tired eyes, late to class.

✓ Adverse Social Behaviour (Why digital media affects people)

TV Violence Research:

- Watching violence for years can make people more aggressive or numb to real violence.

Games:

1. **Violent Content:** Increases aggressive thoughts. (e.g., GTA encourages fighting behaviour)
 2. **Priming Effect:** After playing violent games, people respond faster to angry words or actions.
 3. **Competition:** Even friendly games like FIFA can cause anger or shouting if lost.
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Chapter 4: Intellectual Property Rights

✓ Copyright

- Gives creators ownership of their work.

- Covers: books, music, software, art, movies, broadcasts, etc.
- In Malaysia: 7 types (literary, musical, artistic, film, sound recording, broadcast, derivative).



Example:

Copying someone's song without permission = copyright violation.



Patents

- Protect inventions or new ideas.
- **3 Types:**
 - **Utility:** Machines (e.g., a new phone battery).
 - **Design:** Look/shape (e.g., iPhone design).
 - **Plant:** New plant species.
- **4 Rules (Criteria):**
 - **Novel (New)** – Never made before.
 - **Non-obvious** – Not simple or common.
 - **Useful** – Has purpose.
 - **Lawful** – Not illegal or harmful.
- **Not renewable** (expires after ~20 years) to encourage new ideas.



Example:

Old medicine patents expire → cheaper generic drugs appear.

✓ Software Patent

Harder to get because software = abstract idea (just code).

Problem: Difficult to prove new or useful.

Example: Two apps doing the same function – who owns the idea?

Controversy: Too many patents can block innovation.

✓ Fair Use (4 Factors)

1. **Purpose:** Education/news is okay.
2. **Nature:** Facts more acceptable than art.
3. **Amount:** Using small parts is fine.
4. **Market Effect:** Can't harm the original's sales.

💡 *Example:*

Teacher shows short movie clip in class = fair use.

Uploading full movie on YouTube = not fair use.

✓ Software Protection

- Copyright protects **code, design, documentation**.
- Does **not** protect **ideas or algorithms**.

💡 *Example:*

You can protect how you write the code, but not the idea of “online chat system”.

Chapter 5: Governance of the Internet

✓ Domain Name Hijacking

- Stealing or transferring someone's website name.
Example: Hacker takes "airasia.com" domain.
Protection: Use strong passwords, domain lock, register under ICANN.
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✓ Reverse Domain Name Hijacking

- A company falsely accuses a domain owner to steal the domain.
Example: Small owner of "applezone.com" accused by Apple Inc.
Solution: Report through UDRP, request legal action, claim damages.
- A small web designer registered "applezone.com" in 2005 to showcase his Apple-themed art.
- In 2010, Apple Inc. files a UDRP complaint claiming it violates their trademark.
- The designer proves he owned it before Apple's complaint and used it in good faith.
- ICANN rules it as Reverse Domain Name Hijacking — Apple misused its power.

✓ Typosquatting

- Making fake websites by misspelling names.
Example : "faceboook.com", "goooogle.com".
Used for scams or ads.
Fix: Buy common misspelling domains, enable domain protection.

✓ ICANN (Internet Corporation for Assigned Names and Numbers)

6 Main Roles:

No.	Function	Simple Meaning	Example
1	Domain Name System (DNS) Management	Keeps track of all website names (.com, .org, .my, etc.)	Make sure "google.com" connects to the right IP address.
2	IP Address Allocation	Gives unique Internet addresses (IP numbers) to each device or server.	Your laptop gets its unique IP so it can go online.
3	Protocol Parameter Assignment	Manages technical standards so devices speak the same "Internet language."	Ensures websites, emails, and apps communicate properly.
4	Root Server Management	Controls and maintains the main "root" servers that direct Internet traffic worldwide.	When you type a URL, your request starts at a root server managed by ICANN.
5	Policy Development	Makes global Internet policies (like domain dispute rules).	Example: Rules for Domain Name Disputes (UDRP).
6	Accrediting Registrars	Approves companies that sell domain names.	GoDaddy, Namecheap, and Exabytes are ICANN-accredited registrars.

Model: Multi-stakeholder → includes governments, companies, users.

Importance: Keeps Internet stable and fair for everyone.

Chapter 6: Ethics and Professionalism

✓ Ethical Theories (Simple)

No.	Theory	Simple Meaning	Example (Easy to Remember)
1	Subjective Relativism (主观相对主义) (每个人都有自己的观念)	Everyone has their own opinion about what's right or wrong. There is no universal rule – it depends on each person.	You think cheating in a game is okay because “everyone does it,” but your friend thinks it's wrong. Both think they're right.
2	Cultural Relativism (文化相对主义)	What's right or wrong depends on the culture you live in. No culture's morals are “better.”	In some countries, arranged marriage is normal; in others, it feels unfair. Both follow their own culture's ethics.
3	Divine Command Theory (神圣命令理论)	God decides what is right or wrong. If God commands something, then it's moral.	A person donates to charity because their religion teaches kindness and helping the poor.
4	Ethical Egoism (伦理利己主义)	Do what benefits yourself the most . If it helps you, it's moral.	A programmer hides an easy bug fix so that only they can solve it later and look valuable to the boss.
5	Utilitarianism (功利主义)	An action is good if it gives the greatest happiness to the greatest number of people . Focus on results (consequences) , not intentions.	The government blocks illegal websites to protect millions of users — even if some users are unhappy.
	Act Utilitarianism (行为功利主义)	Judge each single action by its results.	Telling a white lie to cheer someone up.
	Rule Utilitarianism (规则功利主义)	Follow rules that usually bring good results.	Always tell the truth because honesty builds trust.
	Negative Utilitarianism (负面功利主义)	Try to reduce pain or suffering for others.	Provide mental-health breaks to reduce student stress.

	Preference Utilitarianism (偏好功利主义)	Respect people's personal choices or preferences .	A developer makes software with both dark and light modes so everyone is comfortable.
6	Kantianism (康德主义)	Morality depends on duty and intention , not outcome. Do the right thing because it is right , and always treat people as ends, not tools .	You find lost money and return it even though no one will know — because honesty is your duty.
7	Social Contract Theory (社会契约论)	Society works because people agree to follow rules in exchange for safety and order . If you break the rules, you lose that protection.	You obey traffic laws so everyone can drive safely. Government protects your rights if you follow the law.

✓ Software Engineering Code of Ethics

8 Principles:

No.	Principle	Detailed Explanation (with Examples)
1. Public	Software engineers must act in ways that protect the public interest — the safety, privacy, and well-being of users and society come first.	💡 Example: Refusing to release software that could cause financial loss or leak private user data.
2. Client and Employer	Work honestly and responsibly for the company or client, but never ignore public safety . Report problems clearly and avoid conflicts of interest.	💡 Example: If a project collects personal data unsafely, inform the client and recommend a fix.

3. Product	Ensure that software products are safe, reliable, high-quality, and well-tested . Follow proper documentation, testing, and security standards.	💡 Example: Conduct full testing before release; do not skip tests to meet a deadline.
4. Judgment	Be honest and objective in your professional opinions. Avoid bias or decisions influenced by personal gain.	💡 Example: Don't approve faulty software just to please your boss or client.
5. Management	Managers and team leaders must lead ethically , ensure fair work conditions, and enforce ethical standards among their teams.	💡 Example: Assign work based on employee skill levels; don't ask staff to do unsafe or illegal tasks.
6. Profession	Promote the reputation and integrity of the software engineering profession. Support laws and contribute to professional knowledge.	💡 Example: Attend IT ethics workshops; follow data protection laws and promote good practices.
7. Colleagues	Treat coworkers with respect and fairness . Share credit, help each other grow, and don't interfere with others' careers.	💡 Example: Give fair code reviews and help new team members learn best practices.
8. Self	Continuously learn and improve skills. Follow ethical principles personally and professionally at all times.	💡 Example: Stay updated with cybersecurity techniques and avoid cutting ethical corners.

✓ Whistleblowing

When you report wrongdoing in your company.

Before reporting:

1. Serious harm likely
2. Tried internal reporting first
3. Have strong evidence
4. Honest motive
5. Reasonable chance of change

Protection Laws in Malaysia:

- Whistleblower Protection Act 2010
- Employment Act
- Anti-Corruption Act
- Penal Code

💡 Example:

Employee reports company data selling to outsiders → protected by Whistleblower Act.

Chapter 7: Privacy and Surveillance

✓ Types of Privacy

1. **Information:** Control personal data.
2. **Bodily:** Protect physical body (no forced tests).
3. **Communication:** Keep messages private.

4. **Territorial:** Home or personal space privacy.
 5. **Intellectual:** Freedom of thoughts.
 6. **Decisional:** Freedom to make choices (e.g., medical decisions).
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✓ Types of Surveillance

- **Physical:** CCTV, security guards.
- **Digital:** Cookies, GPS, browser tracking.
- **Mass:** Large-scale monitoring (NSA programs).
- **Electronic:** Phone/email monitoring.
- **Biometric:** Face or fingerprint scans.



Example:

Shopping mall uses CCTV → physical surveillance.

Smartphone tracks GPS → digital surveillance.

✓ Eavesdropping (7 Methods)

1. Wiretapping – listening to phone calls.
2. Shoulder surfing – looking at someone's screen.
3. Hidden microphones – bugging rooms.
4. Packet sniffing – capturing network data.
5. Keylogging – recording what you type.
6. Network interception – reading messages.
7. Social engineering – tricking people to reveal secrets.

✓ GDPR vs PDPA

Feature	GDPR (EU)	PDPA (Malaysia)
Scope	All sectors	Private only
Consent	Must be clear	Often implied
Rights	Access, erase, transfer data	Access, correct data
Penalty	Very high (€20M)	RM500,000
Regulator	EU Data Authority	Malaysian PDPA Commissioner

PDPA Personal Data Protection Act 2010

No.	Principle	Explanation (Simplified)	Example
1	General Principle (Consent)	You must get the person's consent before collecting or using their personal data.	Ask user's permission before saving their email.
2	Notice & Choice Principle	Tell users why you're collecting data and give them a choice to opt out.	Provide a privacy notice on website forms.
3	Disclosure Principle	You can only disclose data for the purpose it was collected and with consent.	A company can't sell its customer list to another firm.
4	Security Principle	Take practical steps to protect data from misuse, loss, or leaks.	Use passwords, encryption, and secure storage.

5	Retention Principle	Don't keep personal data longer than necessary.	Delete inactive customer records after 2 years.
6	Data Integrity Principle	Keep data accurate, complete, and up-to-date.	Update customers' contact numbers when notified.
7	Access Principle	Allow individuals to access and correct their data upon request.	Customer can view or edit their personal info on an online portal.

GDPR General Data Protections Regulations

Principle	Meaning (Simplified)	Example
1. Lawfulness, Fairness & Transparency	Data must be collected legally, fairly, and users must be told how it's used.	A company must show a clear privacy policy before collecting user info.
2. Purpose Limitation	Data can only be used for the purpose it was collected for.	Collecting emails for billing → can't use them later for marketing without consent.
3. Data Minimization	Only collect data that is necessary and relevant.	Don't ask for a customer's full address if you only need an email.
4. Accuracy	Data must be kept accurate and up-to-date.	Correct wrong addresses or outdated records.
5. Storage Limitation	Keep personal data only as long as needed.	Delete user info after service ends.
6. Integrity & Confidentiality (Security)	Protect data from unauthorized access, leaks, or damage.	Use encryption and access controls.
7. Accountability	Organizations must prove they follow GDPR rules.	Keep records of data usage and audits.

Chapter 8: Information Age Warfare

✓ Cyberwarfare Types

Type	Meaning	Example
Espionage	Spying using tech	Botnets steal military secrets
Sabotage	Damaging systems	Virus disables power grid
DoS/DDoS	Overloading websites	Attack on government sites
Propaganda	False info online	Fake news to confuse voters
Psychological Warfare	Spreading fear	Fake leaks of citizen data

Chapter 9: Online Crime and Punishment

✓ Social Engineering

- **Phishing:** Fake email to steal passwords.
- **Pretexting:** Pretend to be someone (e.g., bank officer) to get info.
💡 *Example:*
Email says “Click here to reset your bank account” → phishing.

✓ Malware Crimes

Type	Meaning	Example
Virus	Infects files	Corrupts Word docs
Worm	Spreads itself	Sends spam to all contacts
Trojan	Disguised as safe	“Free game.exe” installs malware
Spyware	Watches you	Tracks keystrokes
Ransomware	Locks data for payment	Demands Bitcoin to unlock PC

✓ Malaysia's Cyber Laws

1. Computer Crimes Act 1997 (CCA)

- Section 3–6: Illegal access, modify, misuse.
- Section 7–8: Fraud, damage, data theft.
- Section 9–12: Helping or attempting crime.

2. Communications and Multimedia Act 1998 (CMA)

- Stops spreading fake, offensive, or illegal content.
- Controls service providers and online behaviour.

💡 *Example:*

Posting false info about the government → violates CMA.

Hacking someone's Facebook → violates CCA.

a. Cyber laws of Malaysia (come out)

i. Computer Crime Act 1997

- Criminalizes offenses related to the misuse of computers and digital data.

→ Unauthorized Access (Section 3)

→ Unauthorized Modification of Content (Section 5)

→ Unauthorized Possession of Access Devices (Section 6) 未经授权持有访问设备

→ Unauthorized Use or Disclosure of Access Codes (Section 7) 未经授权使用或披露访问代码

→ Offenses by Officers of a Body Corporate (Section 8) 法人团体官员的违法行为

→ Data Interference (Section 9)

→ Penalties (Section 11)

→ International Cooperation (Section 12)

ii. Communication & Multimedia Act (CMA) 1998

- Regulates the converging communications and multimedia industries.

→ Establishes the MCMC as the regulator.

→ Licensing for service providers.

→ Content regulation (e.g., against offensive or false content).

→ Promotes consumer protection and network security.